

Automated SaaS Onboarding Triage Pipeline

1. Executive Summary

This portfolio submission details the transition of our enterprise SaaS onboarding process from manual JSON payload validation to a fully autonomous, AI-driven triage architecture. By leveraging deterministic large language models within an event-driven orchestration pipeline, the system automatically audits client configurations to identify missing mandatory fields and integration failures. This automated workflow instantly generates standardized, actionable runbooks for deployment teams, drastically reducing time-to-value and eliminating human error in initial client assessments.

2. Operational Problem

The historical reliance on manual parsing of incoming client configuration payloads presents a significant operational and scalability bottleneck during enterprise SaaS onboarding. Requiring technical analysts to manually validate ingestion protocols, authentication keys, and data pipeline connectivity introduces latency and high error rates into the deployment lifecycle. Furthermore, this manual approach severely limits system observability, making it difficult to enforce standardized deployment gating or efficiently track the root causes of integration failures during the critical early stages of client deployments.

3. Technical Architecture

The pipeline is orchestrated via n8n to ensure deterministic, scalable execution through the following stages:

Event-Driven Ingestion: The pipeline initiates via a webhook node configured to execute upon the receipt of POST requests at the `/enterprise-onboarding-triage` endpoint, thereby capturing the initial enterprise configuration JSON payloads.

Agentic Validation: Payload data is routed to an AI Agent powered by the Google Gemini 2.5 Flash model, which operates under a strict deterministic temperature setting of 0.1. Utilizing robust prompt engineering, the agent acts

as an onboarding technical analyst to scan the configuration payload against a block-list, successfully identifying missing mandatory fields and failed data pipeline tests before synthesizing the findings into a structured Markdown format.

Binary Serialization: The workflow automates the conversion of the generated Markdown into a standardized deployment artifact by executing an HTTP POST request to an external microservice (md-to-pdf.fly.dev) via multipart form data, which returns a properly formatted PDF binary.

Dynamic Metadata Handling: To ensure enterprise-grade artifact organization, a custom JavaScript execution node intercepts the converted PDF binary and dynamically assigns a formalized filename by extracting the `company_name` variable directly from the initial webhook payload memory.

4. Operational Runbook: Automated Triage Workflow Execution Overview

This workflow is an event-driven, stateless automation triggered by webhook POST requests. It processes raw client configuration JSON, performs asynchronous AI-driven validation, and exports finalized binary artifacts.

Execution Steps

Stage 1 (Webhook Trigger): Captures configuration payloads via endpoint `/enterprise-onboarding-triage`.

Stage 2 (Agentic Validation): Gemini 2.5 Flash processes the stringified JSON payload to scan for mandatory blockers.

Stage 3 (State Aggregation): The "Edit Fields" node merges raw metadata with audit timestamps for data lineage.

Stage 4 (Binary Serialization): PDF conversion via HTTP-based Markdown rendering.

Stage 5 (Metadata Injection): JavaScript-based dynamic filename assignment using the `company_name` variable.

Outcome Interpretation

Technical Escalation: If the runbook identifies a missing API key or a failed pipeline test (e.g., Vertex Global Logistics), the team must immediately execute the "Critical Blockers" action items.

Launch Readiness: If the runbook identifies no blockers (e.g., Sigma Enterprises), the operations team is cleared to proceed with User Acceptance Testing (UAT).

5. Project Artifacts

Workflow Source: Onboarding Automated Runbook Workflow.txt

Vertex Global Logistics_Runbook.pdf

Sigma Enterprises_Runbook_2.pdf

[Sigma Enterprises_Runbook.pdf](#)

[Onboarding Automated Runbook Workflow.txt](#)

[Vertex Global Logistics_Runbook.pdf](#)